Historical Examples of Statistical Plots

A Course on Data Visualization

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Contents

Overview 1
William Playfair (1759–1823): Bar Charts and Line Charts 1
Florence Nightingale (1820–1910): Polar Area Diagrams 3
John Snow (1813–1858): Cholera Maps 5
Charles Minard (1781–1870): Napoleon Campaign Visualization 6
References 7
Session Info 7

Overview

This document presents four foundational examples in the history of data visualization, featuring pioneering work by William Playfair, Florence Nightingale, John Snow, and Charles Minard. These visualizations established core principles still used in modern statistical graphics.

William Playfair (1759–1823): Bar Charts and Line Charts

Playfair is widely regarded as the inventor of modern statistical graphics. His 1786 *Commercial and Political Atlas* introduced both the line chart and the first known bar chart. [1][2][3]

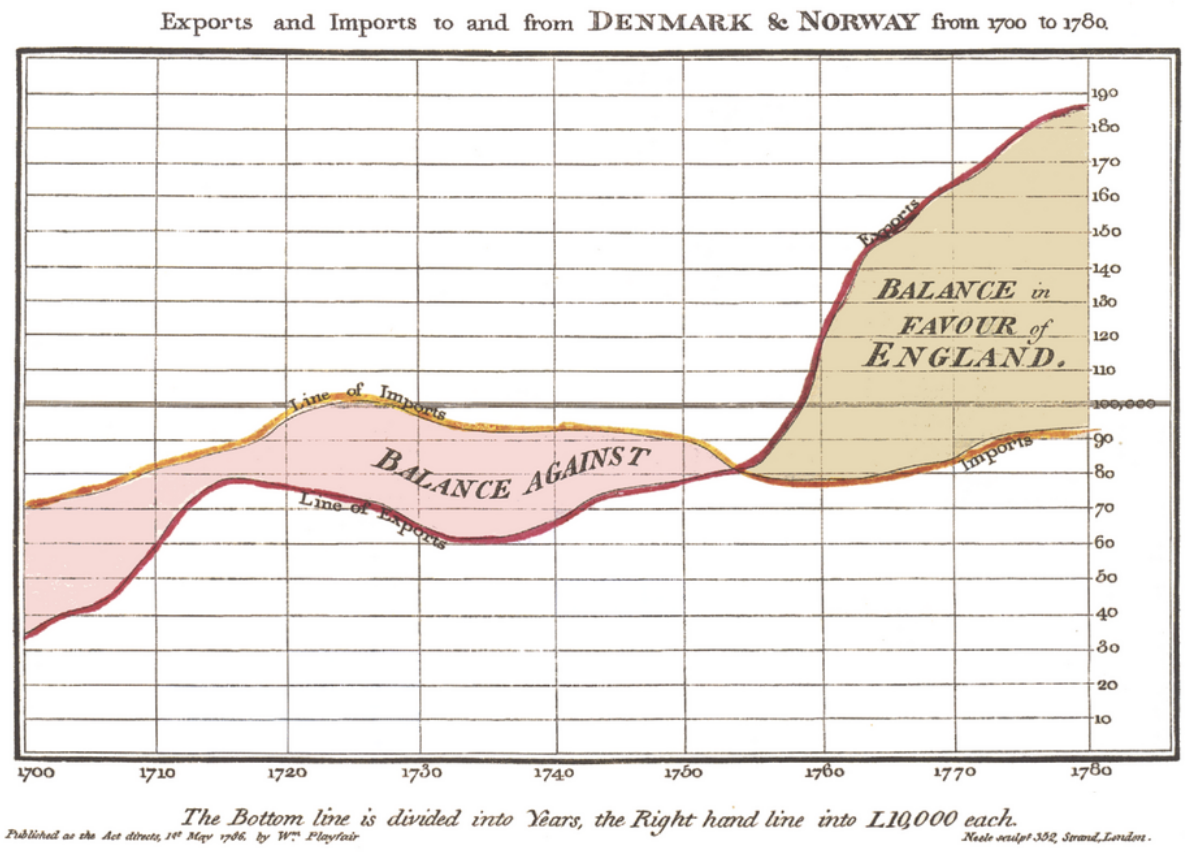
Line Chart Example

(1786) A time-series plot showing England's exports and imports to and from Denmark & Norway from 1700 to 1780. The area between the two lines is shaded to show the balance of trade ("Balance in Favour of England" vs. "Balance Again

[1] History of Information. "William Playfair Founds Statistical Graphics." Available at: https://www.historyofinformation.com/detail.php?entryid=2929

[2] Open Culture. "The Pioneering Data Visualizations of William Playfair." 2023. Available at: https://www.openculture.com/2023/05/the-pioneering-data-visualizations-of-william-playfair-who-invented-the-line-bar-and-pie-charts-circa-1786.html

[3] Wikipedia. "William Playfair." Available at: https://en.wikipedia.org/wiki/William_Playfair



Key features: - Two time series plotted on the same axes - Shaded area between lines to visualize trade balance - X-axis: years (1700–1780) - Y-axis: monetary value (imports/exports)

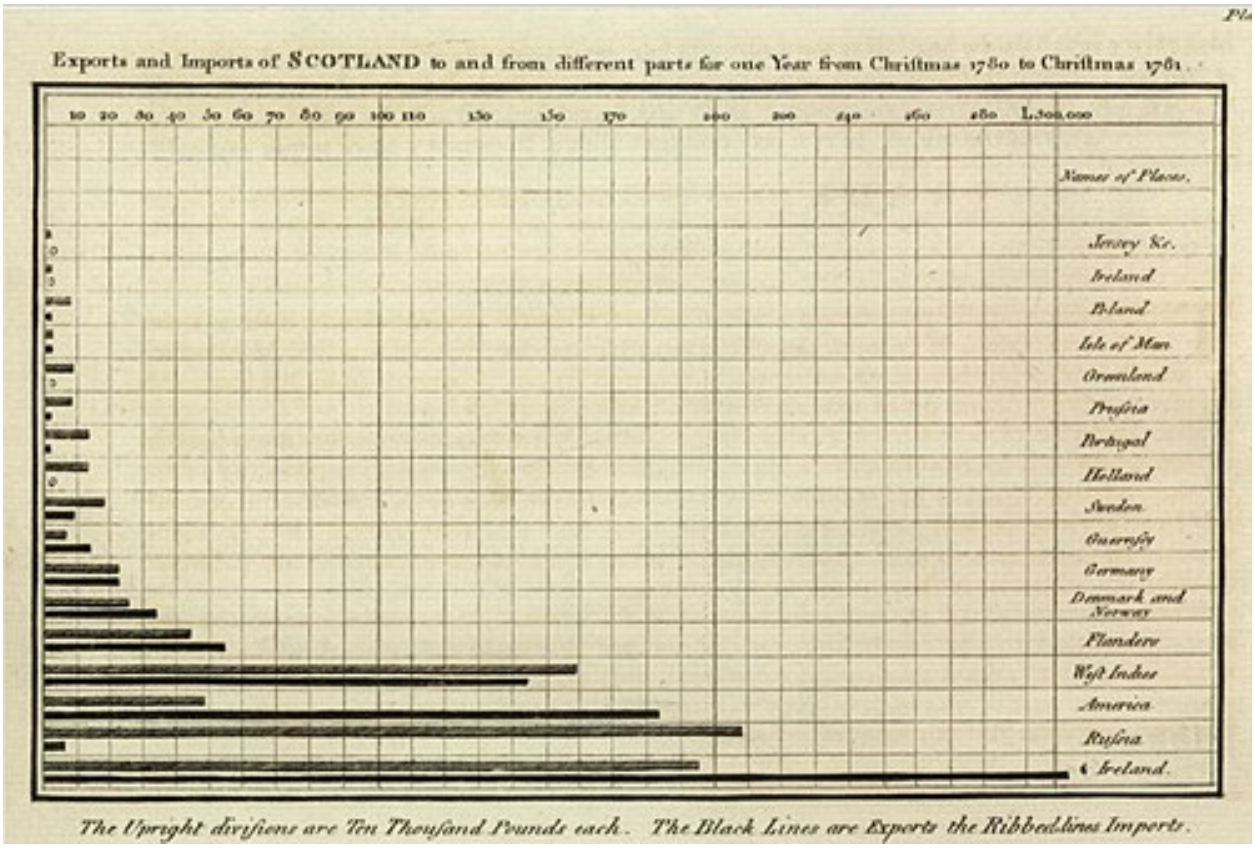
Bar Chart Example

(1786) A horizontal bar chart showing Scotland's exports (solid black bars) and imports (ribbed bars) to and from 17 different trading partners for the year Christmas 1780 to Christmas 1781. This is considered the first quantitative graphical form that did not locate data in space or time. [4][5][6]

[4]History of Information. "William Playfair Founds Statistical Graphics." Available at: https://www.historyofinformation.com/detail.php?entryid=2929

[5]Friendly, M. "The Big Bang: William Playfair, the Father of Modern Graphics." Available at: https://friendly.github.io/HistDataVis/ch05-playfair.html

[6]Conversable Economist. "William Playfair: Inventor of the Bar Graph, Line Graph, and Pie Chart." 2017. Available at: https://conversableeconomist.blogspot.com/2017/08/william-playfair-inventor-of-bar-graph.html



Key features: - Horizontal bars for categorical comparison - Paired bars for exports vs. imports - 17 trading partners on the y-axis - Monetary values on the x-axis

Resources

- Wikipedia: William Playfair
- The Pioneering Data Visualizations of William Playfair
- Gallery of Data Visualization - Historical Milestones
- Lehigh Library Exhibits: Playfair's Pie Chart

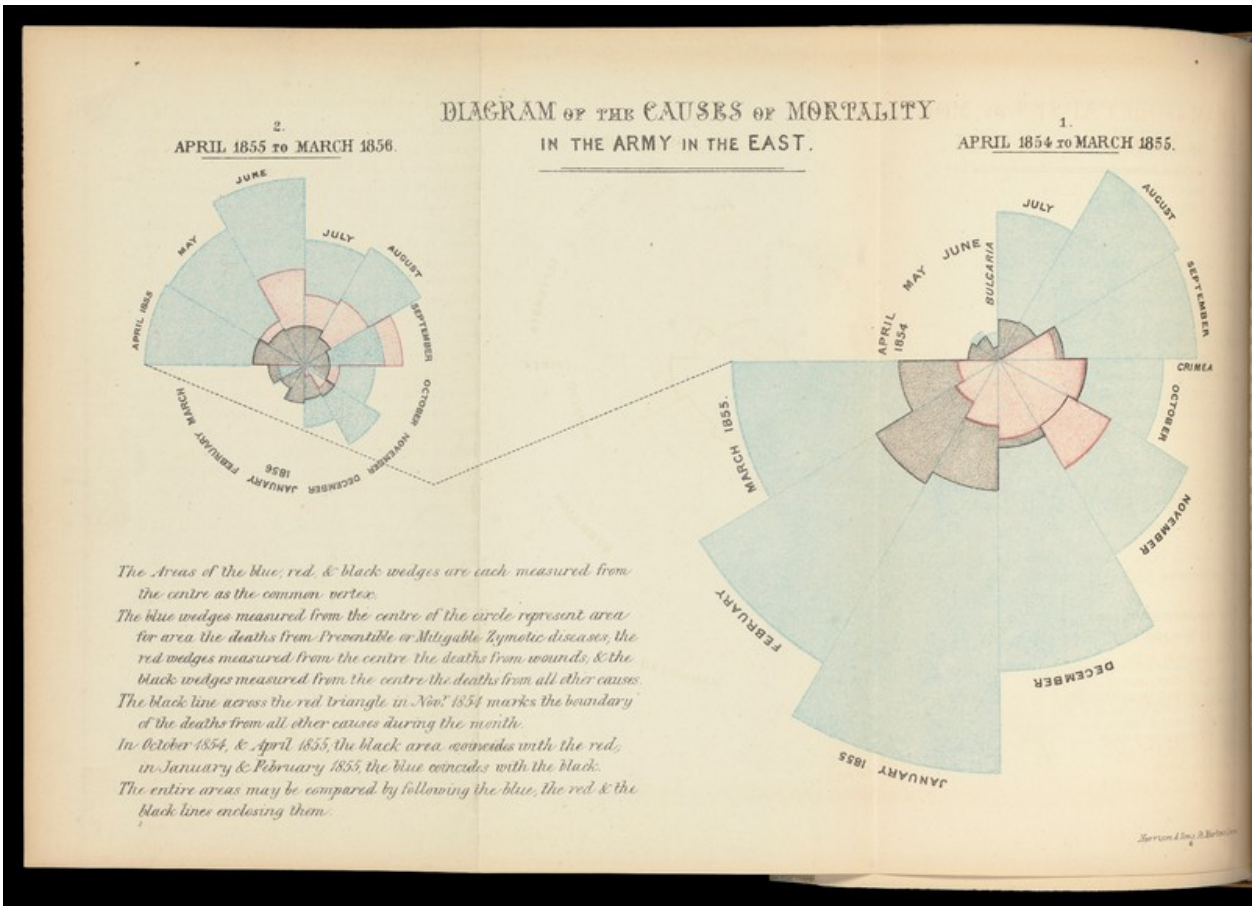
Florence Nightingale (1820–1910): Polar Area Diagrams

Nightingale developed the polar area diagram (also called the "coxcomb" or "Nightingale rose diagram") to communicate mortality data from the Crimean War (1853–56). [7][8][9]

[7] Lehigh Library Exhibits. "Nightingale's Rose Diagram." Available at: https://exhibits.lib.lehigh.edu/exhibits/show/data_visualization/science/nightingale

[8] Wikimedia Commons. "Nightingale Rose Chart." Available at: https://commons.wikimedia.org/wiki/File:Nightingale_Rose_Chart.png

[9] Protovis. "Nightingale's Rose." Available at: https://mbostock.github.io/protovis/ex/crimea-rose.html



Structure

- A circle divided into 12 equal-angle wedges (one per month)
- The **area** of each wedge (not just the radius) is proportional to the mortality rate
- Each wedge is color-coded:
  - **Blue**: deaths from preventable diseases
  - **Red**: deaths from wounds
  - **Black**: deaths from other causes

Purpose

To show that most deaths were from preventable diseases and that mortality dropped dramatically after sanitary reforms were implemented in March 1855. [10][11]

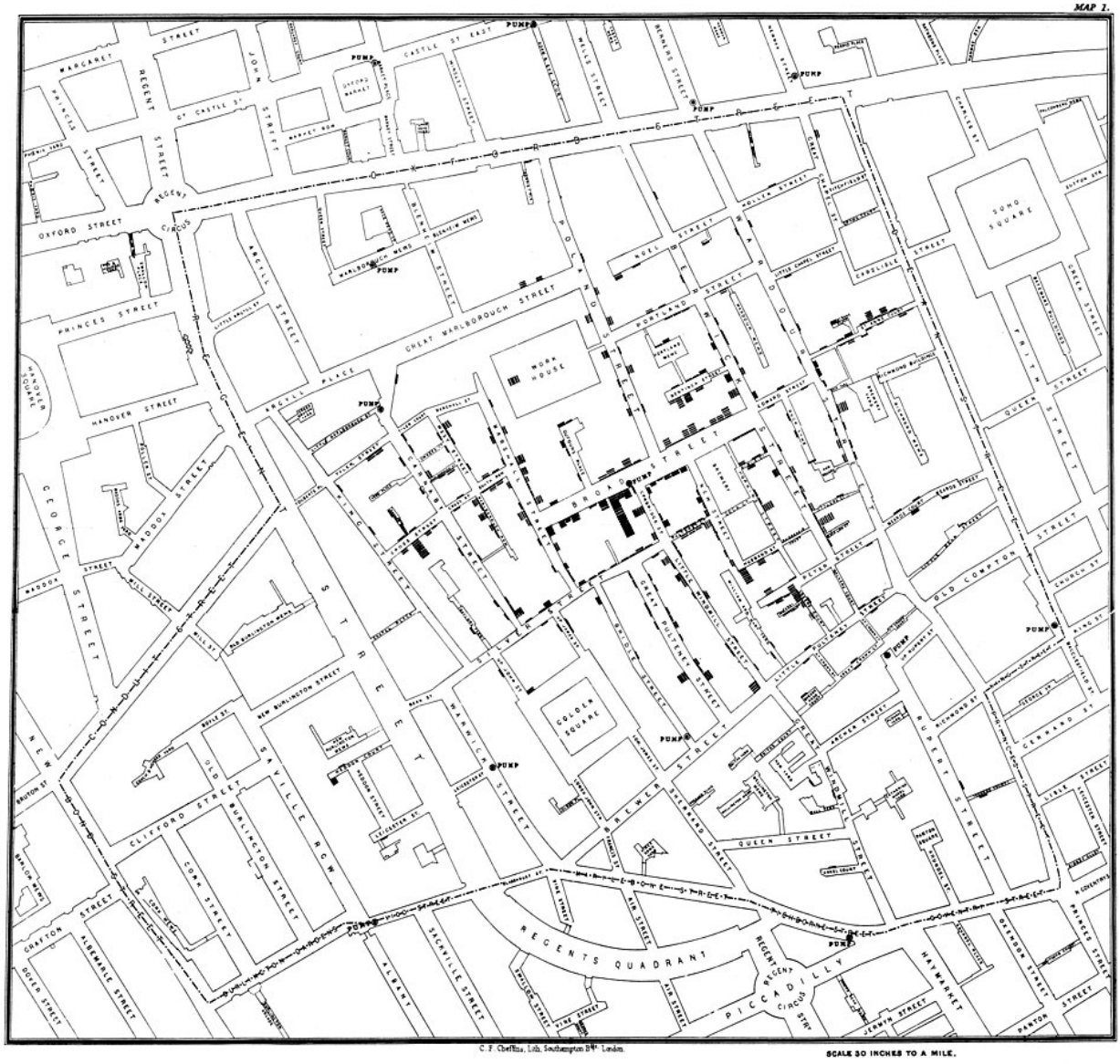
Resources

- Lehigh Library Exhibits: Nightingale's Rose Diagram
- Wikimedia Commons: Nightingale Rose Chart
- Protovis: Nightingale's Rose
- University of St Andrews: Florence Nightingale PDF

[10] Protovis. "Nightingale's Rose." Available at: https://mbostock.github.io/protovis/ex/crimea-rose.html
[11] State of the USA. "Nightingale's Coxcomb Graph." Available at: https://www.stateoftheusa.org/files/nightengale.html

John Snow (1813–1858): Cholera Maps

Snow's 1854 map of the Broad Street cholera outbreak in London's Soho district is a foundational example of spatial epidemiology and data visualization. [12][13][14]



Design

- A street map showing individual cholera deaths as stacked black bars (or rectangles) at each address
- Locations of public water pumps marked as circles
- Focus on the Broad Street pump area

[12] Wikimedia Commons. "Snow Cholera Map." Available at: https://commons.wikimedia.org/wiki/File:Snow-cholera-map-1.jpg

[13] Flourish. "Recreating John Snow's Cholera Map." 2022. Available at: https://flourish.studio/blog/masters-john-snow/

[14] Lehigh Library Exhibits. "Dr. John Snow's Cholera Map." Available at: https://exhibits.lib.lehigh.edu/exhibits/show/data_visualization/science/snow

Insight

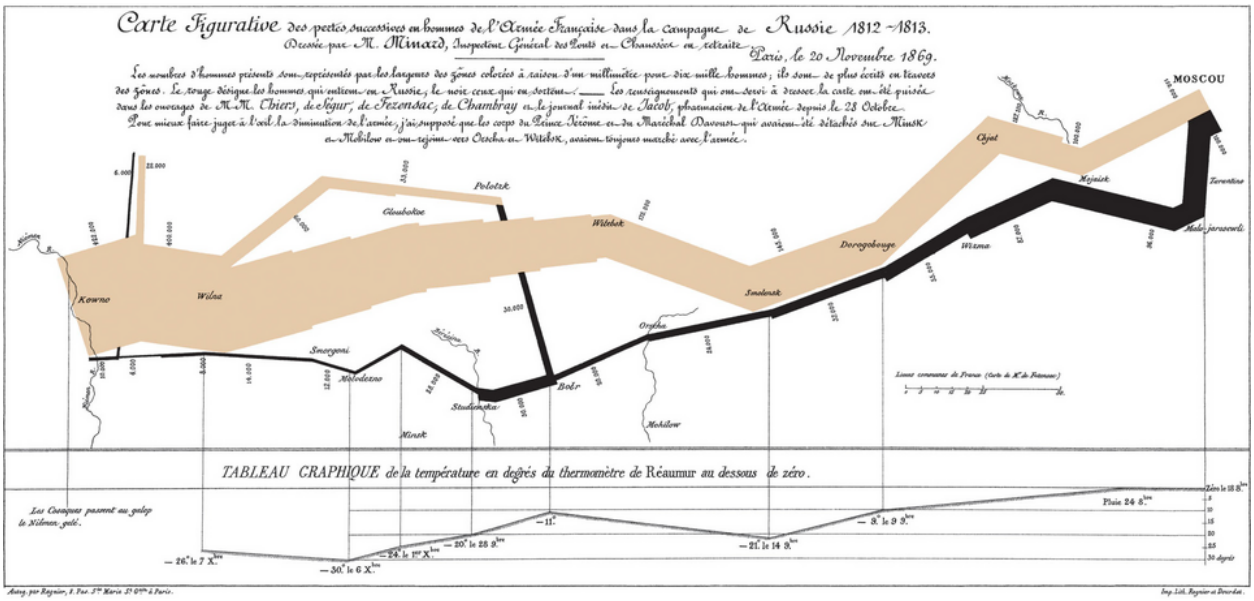
The clustering of deaths around the Broad Street pump provided visual evidence that cholera was waterborne, not airborne (as the miasma theory claimed). This work helped establish the germ theory of disease. [15][16]

Resources

- Wikimedia Commons: Snow Cholera Map
- Flourish: Recreating John Snow's Cholera Map
- ArcGIS: Map a Historic Cholera Outbreak
- Lehigh Library Exhibits: Dr. John Snow's Cholera Map
- The John Snow Archive and Research Companion

Charles Minard (1781–1870): Napoleon Campaign Visualization

Minard's 1869 *Carte figurative des pertes successives en hommes de l'Armee Française dans la campagne de Russie 1812–1813* is often called "the best statistical graphic ever drawn" by Edward Tufte. [17][18][19]



Six Variables in Two Dimensions

Variable	Encoding
Army size	Width of the band (1 mm = 10,000 men), starting at 422,000 and ending at 10,000
Geography	Latitude and longitude (the graphic is a map)
Direction of travel	Color (tan/gold for advance, black for retreat)

[15] Wikimedia Commons. "Snow Cholera Map." Available at: https://commons.wikimedia.org/wiki/File:Snow-cholera-map-1.jpg

[16] Lehigh Library Exhibits. "Dr. John Snow's Cholera Map." Available at: https://exhibits.lib.lehigh.edu/exhibits/show/data_visualization/science/snow

[17] Wikipedia. "Charles Joseph Minard." Available at: https://en.wikipedia.org/wiki/Charles_Joseph_Minard

[18] CMU. "Napoleon's 1812 Campaign." Available at: https://www.cs.cmu.edu/afs/cs/project/sage-1/www/project/samples/sage/Minard-Tufte.html

[19] Masswerk. "Charles Joseph Minard: Napoleon's Retreat From Moscow." Available at: https://www.masswerk.at/minard/

Variable	Encoding
Temperature	Line graph at the bottom shows plummeting temperatures during the retreat
Time	Dates along the retreat path
Location	Relative to key cities, geographic markers, and battle sites

Resources

- Wikipedia: Charles Joseph Minard
- Wikipedia: Minard's Napoleon Map
- CMU: Napoleon's 1812 Campaign
- Masswerk: Interactive Minard Chart
- York University: Visions and Re-Visions of Charles Joseph Minard
- DataViz Blog: Minard's Flow Map

References

Session Info

```
## R version 4.6.1 (2026-06-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=Italian_Italy.utf8  LC_CTYPE=Italian_Italy.utf8
## [3] LC_MONETARY=Italian_Italy.utf8 LC_NUMERIC=C
## [5] LC_TIME=Italian_Italy.utf8
##
## time zone: Europe/Rome
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## loaded via a namespace (and not attached):
##  [1] compiler_4.6.1    fastmap_1.2.0     cli_3.6.6         tools_4.6.1
##  [5] htmltools_0.5.9   otel_0.2.0        rstudioapi_0.19.0 yaml_2.3.12
##  [9] tinytex_0.60      rmarkdown_2.31    knitr_1.51        xfun_0.60
## [13] digest_0.6.39     rlang_1.3.0       evaluate_1.0.5
```

Note on AI assistance

These teaching materials were prepared with assistance from ChatGPT (OpenAI) in drafting explanations, developing examples, and generating R code. The author retains responsibility for the final content.